Task 1:

Q1) ,Q2),Q3),Q4),Q5)

```
user@user:~/Desktop$ ls
12341350
user@user:~/Desktop$ mkdir orion
user@user:~/Desktop$ ls
12341350  orion
user@user:~/Desktop$ cd orion/
user@user:~/Desktop/orion$ ls
user@user:~/Desktop/orion$ mkdir reports scripts
user@user:~/Desktop/orion$ ls
reports  scripts
user@user:~/Desktop/orion$ cd reports/
user@user:~/Desktop/orion/reports$ ls
user@user:~/Desktop/orion/reports$ echo "Project Orion - Confidential" > welcome.txt
user@user:~/Desktop/orion/reports$ ls
welcome.txt
user@user:~/Desktop/orion/reports$ cat welcome.txt
Project Orion - Confidential
user@user:~/Desktop/orion/reports$ ls
welcome.txt
user@user:~/Desktop/orion/reports$ pwd
/home/user/Desktop/orion/reports
user@user:~/Desktop/orion/reports$ pwd > file_location.txt
user@user:~/Desktop/orion/reports$ ls
file_location.txt  welcome.txt
user@user:~/Desktop/orion/reports$ cat file_location.txt
/home/user/Desktop/orion/reports
user@user:~/Desktop/orion/reports$ ls
file_location.txt  welcome.txt
user@user:~/Desktop/orion/reports$ touch notes.txt | cp notes.txt ../scripts/notes_backup.txt
user@user:~/Desktop/orion/reports$ ls
file_location.txt  notes.txt  welcome.txt
user@user:~/Desktop/orion/reports$ cat notes.txt
user@user:~/Desktop/orion/reports$ cat ../scripts/notes_backup.txt
user@user:~/Desktop/orion/reports$ ls ../scripts/
notes_backup.txt
user@user:~/Desktop/orion/reports$ ls -l welcome.txt
-rw-rw-r-- 1 user user 29 Aug 12 15:53 welcome.txt
user@user:~/Desktop/orion/reports$ chmod 600 welcome.txt
user@user:~/Desktop/orion/reports$ ls -l welcome.txt
-rw------- 1 user user 29 Aug 12 15:53 welcome.txt
```

Q6),Q7) Q8)

```
Done.
user@user:~/Desktop/orion/reports$ sudo adduser oriondev
Adding user `oriondev' ...
Adding new group `oriondev' (1001) ...
Adding new user `oriondev' (1001) with group `oriondev' ...
Creating home directory `/home/oriondev' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for oriondev
Enter the new value, or press ENTER for the default
        Full Name []: Oriondev
        Room Number []: 306
        Work Phone []: 123456789
        Home Phone []: 123456789
        Other []: NA
Is the information correct? [Y/n] y
user@user:~/Desktop/orion/reports$ sudo adduser oriondev sudo
Adding user `oriondev' to group `sudo' ...
Adding user oriondev to group sudo
Done.
user@user:~/Desktop/orion/reports$ sudo addgroup orionteam
Adding group `orionteam' (GID 1002) ...
Done.
user@user:~/Desktop/orion/reports$ chown orion orionteam
chown: invalid user: ‘orion’
user@user:~/Desktop/orion/reports$ chown oriondev orionteam
```

Task 2:

Q1)

```
user@user:~$ cd Desktop/
user@user:~/Desktop$ ls
12341350  orion
user@user:~/Desktop$ mkdir task_2
user@user:~/Desktop$ ls
12341350  orion  task_2
user@user:~/Desktop$ cd task_2/
user@user:~/Desktop/task_2$ ls
user@user:~/Desktop/task_2$ cat > employee_attributes.txt << EOF
> Name
> Salary
> Designation
> EOF
user@user:~/Desktop/task_2$ ls
employee_attributes.txt
user@user:~/Desktop/task_2$ cat employee_attributes.txt
Name
Salary
Designation
```

Q2)

```
Designation
user@user:~/Desktop/task_2$ cat > emp1.txt << EOF
> Robert
> 67000
> Manager
> EOF
user@user:~/Desktop/task_2$ ls
emp1.txt  employee_attributes.txt
user@user:~/Desktop/task_2$ cat emp1.txt
Robert
67000
Manager
user@user:~/Desktop/task_2$ cat > emp2.txt << EOF
> Albert
> 45000
> TeamLead
> EOF
user@user:~/Desktop/task_2$ ls
emp1.txt  emp2.txt  employee_attributes.txt
user@user:~/Desktop/task_2$ cat emp2.txt

Albert
45000
TeamLead
user@user:~/Desktop/task_2$ cat > emp2.txt << EOF
> Albert
> 45000
> TeamLead
> EOF
user@user:~/Desktop/task_2$ ls
emp1.txt  emp2.txt  employee_attributes.txt
user@user:~/Desktop/task_2$ cat emp2.txt
Albert
45000
TeamLead
```

Q3)

```
user@user:~/Desktop/task_2$ ls
emp1.txt  emp2.txt  employee_attributes.txt
user@user:~/Desktop/task_2$ paste -s employee_attributes.txt emp1.txt emp2.txt >> emp_data.txt
user@user:~/Desktop/task_2$ cat emp_data.txt
Name    Salary  Designation
Robert  67000   Manager
Albert  45000   TeamLead
user@user:~/Desktop/task_2$ ls
emp1.txt  emp2.txt  emp_data.txt  employee_attributes.txt
```

Q4)

```
user@user:~/Desktop/task_2$ tail -n 2 emp_data.txt >> temp | awk -F" " '{print $1":"$2":"$3}' tem
> temp && mv temp emp_data.txt
user@user:~/Desktop/task_2$ cat emp_data.txt
Robert:67000:Manager
Albert:45000:TeamLead
```

Q5)

```
user@user:~/Desktop/task_2$ tail -n 2 emp_data.txt > temp | awk -F" " '{print $1":"$2":"$3}' temp >
 temp && mv temp emp_data.txt
user@user:~/Desktop/task_2$ cat emp_data.txt
Robert:67000:Manager
Albert:45000:TeamLead
user@user:~/Desktop/task_2$ echo -e "Shelly:28000:SDE\nAntony:27500:SDE\nJazz:29500:SDE\nSuzzy:1500
0:Intern" >> emp_data.txt
user@user:~/Desktop/task_2$ cat emp_data.txt
Robert:67000:Manager
Albert:45000:TeamLead
Shelly:28000:SDE
Antony:27500:SDE
Jazz:29500:SDE
Suzzy:15000:Intern
```

Q6)

```
user@user:~/Desktop/task_2$ awk -F':' '$3=="SDE" {print NR}' emp_data.txt
3
4
5
```

Q7)

```
user@user:~/Desktop/task_2$ grep -i [^s] emp_data.txt
Robert:67000:Manager
Albert:45000:TeamLead
Shelly:28000:SDE
Antony:27500:SDE
Jazz:29500:SDE
Suzzy:15000:Intern
```

Task 3:

Q1)

```
user@user:~/Desktop/task_2$ cd ..
user@user:~/Desktop$ mkdir task_3
user@user:~/Desktop$ ls
12341350  orion  task_2  task_3
user@user:~/Desktop$ cd task__3
bash: cd: task__3: No such file or directory
user@user:~/Desktop$ cd task_3
user@user:~/Desktop/task_3$ ls
user@user:~/Desktop/task_3$ cat > account_details.txt << EOF
> 101,Suresh Kumar,Savings,45000,Delhi
> ^C
user@user:~/Desktop/task_3$ cat > account_details.txt << EOF
> 1001,Suresh Kumar,Savings,45000,Delhi
> 1002,Antaran Sharma,Current,125000,Mumbai
> 1003,Rajiv Singh,Savings,9800,Kolkata
> 1004,Sunita Rao,Current,75250,Chennai
> 1005,Sony Roa,Savings,24000,Bhiali
> 1006,Mohan Das,Savings,60500,Begaluru
> EOF
user@user:~/Desktop/task_3$ cat account_details.txt
1001,Suresh Kumar,Savings,45000,Delhi

1002,Antaran Sharma,Current,125000,Mumbai
1003,Rajiv Singh,Savings,9800,Kolkata
1004,Sunita Rao,Current,75250,Chennai
1005,Sony Roa,Savings,24000,Bhiali
1006,Mohan Das,Savings,60500,Begaluru
user@user:~/Desktop/task_3$ cat > account_details.txt << EOF
1001,Suresh Kumar,Savings,45000,Delhi
1002,Antaran Sharma,Current,125000,Mumbai
1003,Rajiv Singh,Savings,9800,Kolkata
1004,Sunita Rao,Current,75250,Chennai
1005,Sony Roa,Savings,24000,Bhiali
1006,Mohan Das,Savings,60500,Begaluru
EOF
user@user:~/Desktop/task_3$ cat account_details.txt
1001,Suresh Kumar,Savings,45000,Delhi
1002,Antaran Sharma,Current,125000,Mumbai
1003,Rajiv Singh,Savings,9800,Kolkata
1004,Sunita Rao,Current,75250,Chennai
1005,Sony Roa,Savings,24000,Bhiali
1006,Mohan Das,Savings,60500,Begaluru
```

Q2)

```
user@user:~/Desktop/task_3$ awk -F',' '$4>50000 {print $0}' account_details.txt | cut -d, -f1-3 acc
ount_details.txt
1001,Suresh Kumar,Savings
1002,Antaran Sharma,Current
1003,Rajiv Singh,Savings
1004,Sunita Rao,Current
1005,Sony Roa,Savings
1006,Mohan Das,Savings
```